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TURBO DESIGN – OPEN-SOURCE RADIAL EQUILIBRIUM TURBOMACHINERY SOLVER: PART I. TURBINES

Paht Juangphanich[1] Arman Mirhashemi[1] William Andress[2]
[1]NASA Glenn Research Center, Cleveland OH
[2]Purdue University, West Lafayette IN

ABSTRACT

Advances in 3D Geometrical Designs and Cooling have played a significant role in improving the efficiency of turbomachinery. However, these advancements must be effectively translated back to the modeler. Machine learning can facilitate this transition. Specifically, machine learning–based loss models can bridge the gap between 3D and 1D designs, enabling modelers not only to predict velocity triangles but also to extract additional geometric features. Currently, the design tools used at NASA have not been updated to support such integration—until now. Turbo-Design is an open-source, Python-based tool that replaces TD2 (LEW-11029-1) and AXOD2 (LEW-16323-1), both of which are radial equilibrium solvers for axial turbines. The goal of this update is to enable the integration of machine learning loss models into radial equilibrium equations. Additionally, Turbo-Design is designed to support radial machines. This paper presents the governing equations, the underlying assumptions, the integration of legacy loss models, a framework for future integration of machine learning models, and a validation comparison with CFD.

Keywords: Radial Equilibrium, Loss Models, Turbines, Compressors, Design

NOMENCLATURE

Roman Letters

P	Pressure [Pa]
R	Gas constant [J/(kg·K)]
T	Temperature [K]
V	Absolute velocity [m/s]
W	Relative velocity [m/s]
U	Blade velocity [m/s]
H	Specific enthalpy [J/kg]
I	Rothalpy [J/kg]
Cp	Specific heat at constant pressure [J/(kg·K)]
r_M	Meridional radius of curvature [m]
c	chord [m]
c_p	specific heat at constant pressure [J/(kg·K)]
f	objective function [-]
h	specific enthalpy [J/kg]
s	Specific entropy [J/(kg·K)]
v	specific volume [m³/kg]
w	specific work [J/kg]
$\dot{m}$	mass flow rate [kg/s]

Dimensionless Groups

rp	Degree of Reaction
Yp	Pressure loss coefficient [-]
Y+	Dimensionless wall distance [-]
ψ	stage loading

Greek Letters

α	absolute flow angle [deg]
β	relative flow angle [deg]
γ	Ratio of specific heats [-]
η	efficiency
σ	standard deviation
ϕ	angle of inclination
$\hat{\imath}$	unit vector; subscript denotes direction (r, θ, M, ax)

Subscripts

0	Total Condition
ax	axial
M	meridional
metal	metal angle
TT	Total-Total
TS	Total-Static
R	Relative frame
θ	Tangential component

INTRODUCTION

The foundation of turbomachinery design lies in 1D mean-line analysis, which simplifies the radial equilibrium equations by assuming a constant mean radius. This simplification is necessary to obtain rapid estimates of velocity triangles. However, the actual flow within compressors and turbines is not truly one-dimensional; there is a radial component to the velocity that shifts streamlines and redistributes mass flow.

Radial equilibrium programs provide a higher-fidelity analysis of flow paths by accounting for these variations. Historically, NASA has released several such tools, including the Axial Turbine Design and Performance Code (AXOD) [1], Turbine Design 2 (TD2) [2], and the Radial-Inflow Turbine Conceptual Design Code (RTD) [3]. TD2 is used to generate designs at design conditions, while AXOD evaluates axial turbines under off-design conditions. All of these solvers were developed between the 1970s and 1990s in Fortran 77. Unfortunately, the theories and assumptions behind these tools are either poorly documented or have been lost over time, making it difficult to verify results or implement modifications. As a result, commercial codes have become more attractive alternatives.

Commercial tools such as AxSTREAM [4] by SoftInWay and AXIAL [5] by NREC also use radial equilibrium–based streamline analysis to design compressors and turbines. Some, like SoftInWay, have begun incorporating machine learning into their codes [6]. However, the proprietary nature of these tools makes it challenging for academic and government institutions to implement custom AI inference models or to modify the code to include additional physics.

Loss modeling plays a critical role in generating accurate velocity triangles. Most loss models in turbomachinery are derived from wind tunnel experiments using cascades. However, these datasets are often incomplete or based on outdated, suboptimal designs. For instance, Ainley and Mathieson [7] estimated losses in axial steam turbines during the 1940s and 1950s, while Traupel [8] published loss correlations in 1966—though his work is not easily accessible. The well-known Craig and Cox [9] model includes necessary data for predicting profile loss, but lacks detail on secondary flow losses. Their model does, however, account for effects such as cavities and windage.

Modern turbomachinery designs feature optimized geometries that significantly reduce losses. Incorporating these advancements into streamline-based analyses requires flexible, extensible tools. This paper documents the solution methodology and assumptions underlying Turbo Design, an open-source Python framework that enables integration of custom loss models, including machine learning based approaches.

SOLUTION METHODOLOGY

Coordinate System

The coordinate system is defined in Figure 1. The angle between the absolute velocity V and the meridional velocity V_M is referred to as the absolute flow angle α. In most textbooks, this angle is typically defined relative to the axial velocity V_{ax}; however, that definition is not suitable for radial machines or turbomachinery with flow paths that vary radially. Equations (1 - 4) express all velocity components in terms of the meridional velocity.

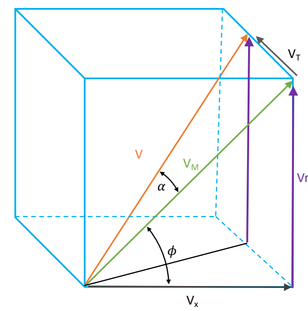

Figure 1. Coordinate System used to derive the rest of the equations. Note: α is between V and V_M; ϕ is between V_M and V_{ax}

$$V_{ax} = V_M \cos(\phi) \tag{1}$$

$$V_r = V_M \sin(\phi) \tag{2}$$

$$V_T = V_{ax} \tan(\alpha) = V_M \tan(\alpha) \tag{3}$$

$$V^2 = V_{ax}^2 + V_r^2 + V_T^2$$
$$V^2 = V_M^2(1 + \tan^2(\alpha)) \tag{4}$$

Streamline locations are defined as a percentage from hub to shroud.

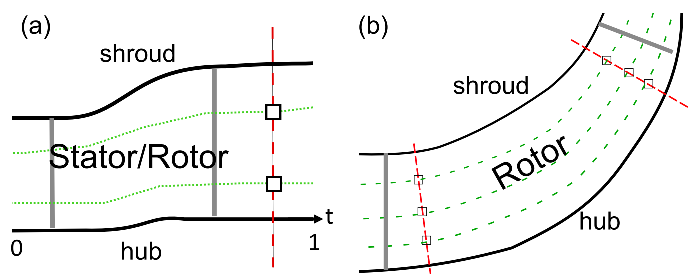

Figure 2. Example of Streamline Definition

Factoring in Curvature

Turbo Design can accommodate channels defined either axially or radially, as shown in parts (a) and (b) of Figure 3. This flexibility is possible because both x and r are expressed as functions of the nondimensional variable t, which represents the percentage of length along the hub. Earlier versions of the code (TD2 and AXOD) supported only jagged flowpath geometries, where the row exit points are known, as illustrated in Fig. 3a. While such an approach may be acceptable for rough calculations, detailed channel geometry is essential for accurate curvature calculations, which in turn affect turning angles and streamline positions.

Smooth channel curves better represent actual designs and lead to more accurate predictions of the radius of curvature r_M and inclination angle ϕ, both of which are critical for radial equilibrium analysis. The first step in defining the geometry is to specify the points along the hub and shroud that define the channel. Blade rows are then inserted, with axial positions defined as a percentage of the hub length. A dashed line is drawn perpendicular to the hub to intersect the shroud. In axial machines, this is replaced with a line of constant radius. Figure 2 illustrates the difference in cut-plane definitions (dashed lines) between axial and radial machines.

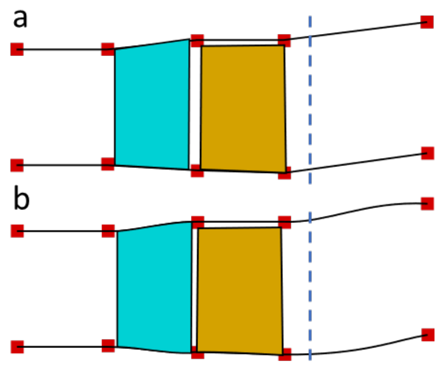

Figure 3. a) Jagged design. b) Smooth Channel Design

Radial Equilibrium

Radial equilibrium describes the balance between pressure forces and inertial forces acting on a fluid element along the blade span of a compressor or turbine passage. Radial flow arises primarily from two sources: (1) the radial component of velocity due to changes in radius from the first to the last stage, and (2) the pressure force distribution required to maintain force balance in the radial direction.

As the flow exits a stator or rotor row, it acquires a whirl (tangential) velocity component (V_T), which generates a centripetal force on the fluid. To counteract this, the static pressure must increase with radius—i.e., pressure rises radially as tangential velocity increases. Any rise in pressure force is a result of linear acceleration of the fluid. This relationship is key to maintaining radial equilibrium.

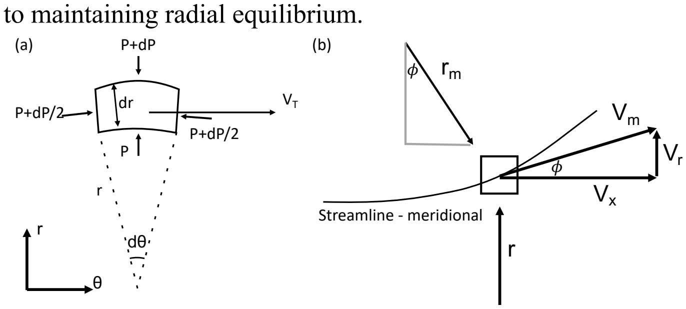

Figure 4. Radial Equilibrium of Fluid Element in *r-θ* (a) and *r-x* (b) planes

Figure 4a illustrates the balance of forces in the *r-θ* plane. Two primary forces act on the fluid element: the net pressure force ($F_{P,net}$) and the centripetal force (F_c). The sign convention used is negative for radial outward and positive for radially inward centripetal force. Fig. 4b shows a similar fluid element in the *r-x* plane, where two additional forces are present: the meridional radial force ($F_{M,r}$) and the meridional centripetal force ($F_{M,c}$). The sum of all forces in the radial direction is expressed in (Eq 5).

$$\sum \vec{F} = \vec{F}_{P,net} = (F_c + F_{M,c} + F_{M,r})\hat{\imath}_r \quad (5)$$

Net Pressure force ($F_{P,net}$) is the sum of all pressure forces on a fluid element in the r-θ frame: top, bottom, and side faces. This equals the summation of centripetal and meridional forces. The top and bottom forces are defined as:

$$\vec{F}_{top} = (P + dP)(r + dr)d\theta\hat{\imath}_r$$

And,

$$\vec{F}_{bottom} = -Prd\theta\hat{\imath}_r$$

The pressure force on the two side faces of the wedge are defined at angles θ and $\theta+d$ is the average pressure $P_{avg} = P + \frac{dP}{2}$. The key to deriving the last part is the unit tangent vector which rotates as you move in the θ direction.

$$(\theta + d\theta)\hat{\imath}_\theta = \hat{\imath}_\theta - d\theta\hat{\imath}_r$$

The force on face θ is:

$$\vec{F}_\theta = P_{avg}dr\,\hat{\imath}_\theta$$

The force on face $\theta+d\theta$ is therefore:

$$\vec{F}_{\theta+d\theta} = -P_{avg}dr(\theta + d\theta)\hat{\imath}_\theta$$

Summing $F_\theta + F_{\theta+d\theta}$ cancels out $\hat{\imath}_\theta$term which results in:

$$\vec{F}_{sides} = -\left(P + \frac{dP}{2}\right)drd\theta\hat{\imath}_r$$

$\vec{F}_{sides}$ points radially outward but according to the sign convention used in this paper it should be negative hence the negative sign in front. (Eq 6) is the sum of forces on the top, bottom and sides.

$$\begin{aligned}\vec{F}_{P,net} &= \vec{F}_{top} + \vec{F}_{bottom} + \vec{F}_{sides} \\ &= \left((P + dP)(r + dr)d\theta - Prd\theta - \left(P + \frac{dP}{2}\right)drd\theta\right)\hat{\imath}_r\end{aligned} \quad (6)$$

Neglecting higher order terms, the equation simplifies to:

$$\vec{F}_{P,net} = rdPd\theta\hat{\imath}_r \quad (7)$$

Centripetal force is the force needed for the particle to maintain its direction in the streamline.

$$F_c = \frac{\rho r V_T^2}{r}drd\theta = \rho V_T^2 drd\theta\hat{\imath}_r \quad (8)$$

There are two meridional forces in the *r-x* plane (Figure 4), for a streamline that is curved: The centripetal force ($F_{M,c}$) keeping the particle along the streamline and the radial component of force ($F_{M,r}$) required to keep the particle attached to the streamline. The centripetal force is a function of the meridional velocity (V_M), the radius of curvature (r_M) and the angle of inclination (ϕ).

$$F_{M,c} = ma = -\frac{mV_M^2\cos(\phi)}{r_M}\hat{\imath}_r \quad (9)$$

Substituting $m=\rho rdrd\theta$ reduces to:

$$F_{M,c} = -\frac{V_M^2}{r_M} cos(\phi)\, \rho rdrd\theta \hat{\imath}_r \quad (10)$$

The radial component of force in the meridional direction required to create linear acceleration along the streamline is defined in (Eq 11):

$$\vec{F}_{M,r} = -\frac{dV_M}{dt} sin(\phi)\, \rho rdrd\theta \hat{\imath}_r \quad (11)$$

Using the property from material derivatives:

$$\frac{dV_M}{dt} = \frac{dM}{dt}\frac{dV_M}{dM} = V_M \frac{dV_M}{dM} \hat{\imath}_r \quad (12)$$

This changes (Eq 11) to:

$$\begin{aligned} \vec{F}_{M,r} &= -V_M\, sin(\phi) \frac{dV_M}{dM} \rho rdrd\theta \hat{\imath}_r \\ &= -V_r \frac{dV_M}{dM} \rho rdrd\theta \hat{\imath}_r \end{aligned} \quad (13)$$

The term $\frac{dV_M}{dM}$ in (Eq 13) is negligible for axial machines but is retained here for radial machines. Because the meridional direction is neither purely axial nor purely radial, the gradient is evaluated by decomposing a differential step along the streamline into its axial and radial components using the unit vectors, $\hat{\imath}_M dM = \hat{\imath}_x dx + \hat{\imath}_r dr$, where $dM = \sqrt{(dx)^2 + (dr)^2}$. Projecting the velocity gradient onto both directions, rather than the axial direction alone as in axial-only solvers, is what preserves the radial-curvature contribution to (Eq 15).

Substituting (Eqs 7, 8, 11, 13) into (Eq 5) produces in the $\hat{\imath}_r$ direction:

$$\begin{aligned} rdPd\theta\hat{\imath}_r = (&\rho V_T^2 drd\theta - \rho rdrd\theta \frac{V_M^2}{r_M} cos(\phi) \\ &- V_r \frac{dV_M}{dM} \rho rdrd\theta)\, \hat{\imath}_r \end{aligned} \quad (14)$$

Which is the radial equilibrium that appears in OTAC User Manual [10] and in Cumpsty [11]:

$$\hat{\imath}_r:\ \frac{1}{\rho}\frac{dP}{dr} = \frac{V_T^2}{r} - \frac{V_M^2}{r_M} cos(\phi) - V_r \frac{dV_M}{dM} \quad (15)$$

Appendix A contains additional details on the solution procedure for (Eq 15).

Assumptions and Limitations

The radial equilibrium formulation in Turbo Design rests on several key assumptions that define its range of applicability. The flow is assumed to be steady, axisymmetric, and adiabatic (except where cooling is explicitly modeled). The working fluid is treated as a calorically perfect gas with constant specific heat ratio (γ) and specific heat at constant pressure (Cp), though variable properties can be incorporated through Cantera for air. Rothalpy is conserved across each blade row, implying ideal (lossless) stator passages unless explicit loss models are applied. Blade lean effects are assumed negligible, meaning the blade-to-blade pressure gradient is small compared to the hub-to-tip gradient.

These assumptions impose practical limitations on the solver's applicability. As discussed in the Solution Stability section, the C parameter (Eq 35, Appendix A) must remain below unity to avoid numerical instabilities associated with choked throughflow. Validation cases on GitHub demonstrate solver performance with relative Mach numbers exceeding unity ($M_{rel} > 1$), representative of highly loaded transonic turbines. The solver does not capture unsteady phenomena such as blade row interactions, rotating stall, or surge. Secondary flow effects (tip leakage, hub corner separation, passage vortices) are not explicitly modeled but can be incorporated through appropriately calibrated loss correlations.

Solver Modes

There are two modes for solving these equations: Pressure Balance and Angle Matching. Pressure Balance mode assumes blade angles are already defined and is applicable for both design and off-design analysis. The solver balances pressure between blade rows simultaneously using SciPy's [12] SLSQP (Sequential Least Squares Programming) optimizer. To improve convergence, pressure is expressed as fractional drops between blade rows, each bounded between 0 and 1. For example, a 4-row turbine (2 stages) with Total Pressure defined at inlet and Static Pressure at exit requires 3 variables. A solution of [0.3, 0.3, 0.4] indicates 30% of the total pressure drop occurs across the first stator, 30% across the first rotor, and 40% across the second stator, with the remainder (rotor exit) matching the outlet static pressure. This formulation ensures monotonic pressure decrease through the stage and improves convergence.

In Angle Matching mode, the blade angles are unknown. A mass flow distribution target can be specified from hub to shroud. The solver iterates through each blade row to find the flow angles that satisfy this distribution. Adjusting the mass flow distribution can be used to control the blade twist – variation of blade exit angle.

For multi-stage turbines, the solver minimizes an objective function that balances mass flow across all blade rows:

$$f = 2\sqrt{\frac{1}{n}\sum_{i=1}^{n}(\dot{m}_i - \bar{\dot{m}})^2} = 2\sigma(\dot{m}) \quad (16)$$

where σ denotes the standard deviation function and $\dot{m}_i$ is the mass flow rate (excluding coolant) at blade row *i*. The factor of two amplifies deviations to improve optimizer sensitivity. This formulation drives the optimizer toward solutions where mass is conserved throughout the turbine while maintaining correct pressure drops through each blade row.

Solution Stability

The derivation in the Appendix reveals an important parameter for solution stability. The *C* parameter represents the ratio of kinetic energy to total enthalpy, which physically corresponds to the local Mach number squared scaled by thermodynamic properties. When C approaches or exceeds unity, the flow becomes choked—the local velocity reaches the sonic limit, and no additional mass can pass through the passage. Mathematically, this causes the term (1-C) in the denominator to approach zero, producing numerical instabilities (NaN values).

Common causes of solver instability include: (1) specifying a mass flow rate that exceeds the choked flow limit for the given geometry; (2) excessive flow turning angles that constrain the effective passage area; and (3) inlet conditions that place the flow near sonic conditions. These conditions are detected automatically, and diagnostic messages suggest corrective actions such as reducing the target mass flow, decreasing turning angles, or adjusting inlet conditions. This physical interpretation of the C parameter defines the solver's operating envelope: C must remain below unity to avoid numerical instabilities associated with choked throughflow.

Adjusting Streamlines

Streamline locations can be automatically adjusted to balance the mass flow distribution from hub to shroud. Adjusting the streamlines affects key variables: r_M, ϕ, V_M, P_0. Because of this interdependence, a convergence loop is required to iteratively adjust and solve for consistent flow properties.

Addition of Cooling

When factoring in cooling for gas turbines, the total temperature at the stator or rotor exit is affected. Calculation of exit total temperature can be done using the mass-flow-Cp weighted approach.

$$T_0 = \frac{\dot{m}CpT_0 + \dot{m}_cCp_cT_{0c}}{\dot{m}Cp + \dot{m}_cCp_c} \tag{17}$$

When it comes to the rotor, the relative total temperature is affected by the introduction of cooling. The absolute total temperature is back-calculated assuming the velocity triangles do not change. (Eqs 17-20) describe the procedure for computing total temperature at the rotor exit. In (Eq 19), W denotes the relative velocity (velocity in the rotating frame of reference) and U denotes the blade rotational velocity. The addition of cooling affects the thermodynamic properties of T_0 which has an effect on the static temperature and velocity triangles, but it does not take into account any aerodynamic interactions that would also alter the triangles.

$$T_{0R} = \frac{\dot{m}CpT_{0R} + \dot{m}_cCp_cT_{0c}}{\dot{m}Cp + \dot{m}_cCp_c} \tag{18}$$

$$T = T_{0R} - \frac{W^2 - U^2}{2Cp} \tag{19}$$

$$T_0 = T + \frac{V^2}{2Cp} \tag{20}$$

(Eq 21) is the Rothalpy equation [13], it describes the relationship between total enthalpy with the rotational energy. Conservation of Rothalpy is assumed across all blade rows.

$$I = H_{0R} - \frac{U^2}{2} \tag{21}$$

Turbine Legacy Loss Models

Authors of turbine loss models have developed equations that relate geometric parameters—such as axial chord, overall chord length, camber (or backbone), leading and trailing edge angles, and pitch-to-chord ratio—to performance metrics like pressure loss or enthalpy loss [7–9, 14]. However, integrating these models into an optimization presents several challenges.

Many of these equations require coefficients that must be extracted from printed figures. This process is inefficient, especially since older publications often present figures at skewed angles, making interpolation difficult. Furthermore, the original works typically do not report the accuracy of their curve fits, adding uncertainty to the predictions.

A practical solution is to digitize the figure data and fit a smooth mathematical surface to it. The Turbo Design GitHub repository provides CSV datasets for several widely used models, including Ainley–Mathieson [7], Kacker–Okapuu [14], Craig–Cox [9], and Traupel [8]. These datasets have been fitted using B-Spline surfaces. For example, Fig. 5 shows a B-Spline surface fitted to data that includes leading-edge flow angle (β_1), trailing-edge flow angle (β_2), and stagger angle. In this case, the input values (β_1 and β_2) are used to predict the stagger angle.

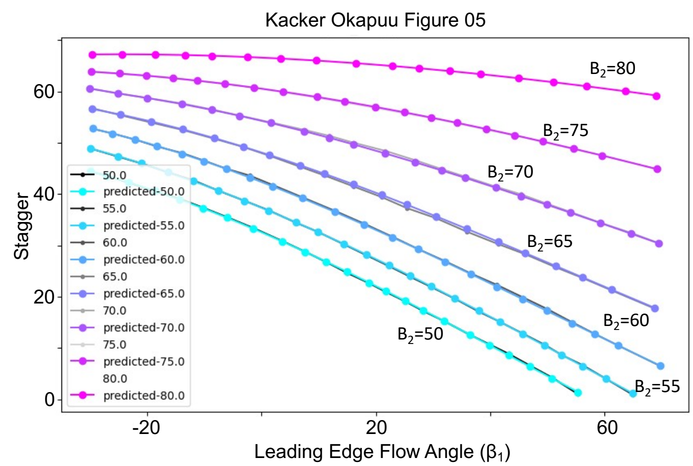

Figure 5. Kacker Okapuu B-Spline surface fit example.

Enthalpy, Pressure and Entropy Loss

Many legacy turbine loss correlations are based on enthalpy or total pressure loss, while others—such as Denton's model [15]—are formulated using entropy loss. When defining either pressure or entropy loss correlations, the temperature component can be inferred from the T–s diagram. For a stator, where no work is done, the total temperature remains constant ($T_{01}=T_{02}$). Similarly, in the rotor (in the relative frame), no work is done, so $T_{02R} = T_{03R}$. Since pressure and entropy are thermodynamically linked, a change in one implies a change in the other. Figure 6 illustrates how entropy loss and total pressure loss manifest in the flow.

Enthalpy-based loss correlations were developed to estimate overall stage efficiency. However, a known enthalpy or total temperature drop does not uniquely determine the corresponding total pressure, as a range of total pressures can produce the same total temperature drop. This ambiguity is illustrated in Figure 7.

In this framework, each stator–rotor pair is treated as a single combined stage. The total temperature drop across the stage is used to estimate the total-to-total efficiency. A single-objective optimization loop then adjusts the pressure loss to match the efficiency predicted by the enthalpy-based loss model. Subsequently, all flow quantities are resolved using the radial equilibrium equations. However, the accuracy of these solutions may be affected by the limitations inherent in enthalpy-based loss modeling.

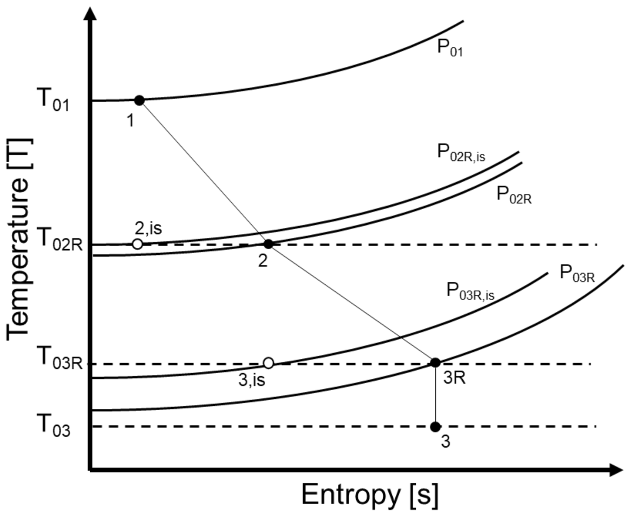

Figure 6. Turbine T-s Diagram for Pressure or Entropy Loss

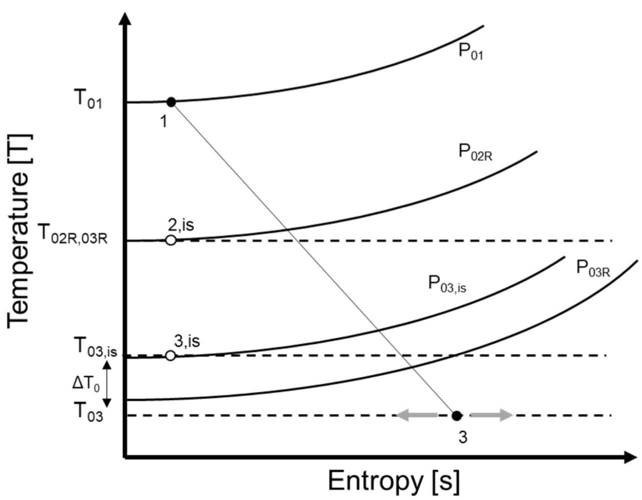

Figure 7. Turbine TS Diagram for Enthalpy Loss

Defining a Framework for Machine Learning Loss Models

Each engine and testing facility has unique characteristics, including specific disk windage, leakage losses, etc. Consequently, loss models developed from experimental data or computational simulations are often tuned to specific configurations. To improve generalizability, the authors propose using machine learning to generate custom loss models.

To support this, the loss model architecture is designed with a modular interface: each loss model inherits from a common base class that receives both upstream and current blade row properties, enabling straightforward substitution of legacy correlations with trained inference models. The ML models are primarily intended to interpolate within a known design space defined by the training data. Extrapolation to configurations outside the training envelope is not recommended without additional validation, as neural network predictions can degrade significantly beyond the training domain. Validation of ML-based loss models is deferred to future work, where the authors plan to demonstrate trained models with performance metrics against established experimental datasets.

Off-design conditions can be incorporated by evaluating incidence angle—the difference between blade metal angle and incoming flow angle. This value can be passed to the loss model. The Craig–Cox model, demonstrated in the GitHub tutorial, accounts for incidence effects in its formulation.

Efficiency Definitions

Two efficiency definitions are commonly used in turbine analysis. Total-to-total efficiency compares the actual work extracted to the isentropic work at the same exit total pressure, appropriate when the exhaust kinetic energy is recovered. Total-to-static efficiency uses exit static conditions as the reference, appropriate when exhaust kinetic energy is not recovered. The standard total-to-total efficiency formula (Eq 27a) uses absolute total temperatures:

$$\eta_{TT} = \frac{T_{01} - T_{02}}{T_{01} - T_{02,is}} \tag{27a}$$

For radial turbines with significant radius change, (Eq 27a) can produce misleading results because the absolute total pressure ratio is dominated by frame-change effects (centrifugal work) rather than aerodynamic losses, yielding efficiency values near unity even with significant irreversibilities. Turbo Design addresses this by implementing an entropy-based efficiency (Eq 27b):

$$\eta_{TT} = \frac{w}{w + T_2 \Delta s} \tag{27b}$$

Where $w = c_p(T_{01} - T_{02})$ is the actual specific work, $T_2$is the exit static temperature, and Δs is the entropy generation. By applying the Gibbs equation in the rotating frame where relative total enthalpy is conserved (see Appendix A, Eqs 28a-c and 29a-d), Δs simplifies to (Eq 27c):

$$\Delta s = R ln\left(\frac{P_{0R,in}}{P_{0R,out}}\right) \tag{27c}$$

Where P_{0R} denotes relative total pressure. The only mechanisms that reduces P_{0R} in the rotating frame are irreversibilities such as friction, separation, shock losses, and wake mixing without contamination from frame-change effects. Consistency with the

Second Law is enforced by requiring $\Delta s \geq 0$, and the formulation falls back to (Eq 27a) when relative total pressure data is unavailable.

Power is computed two ways for verification: Euler power from velocity triangles as $Euler\ Power = \dot{m}(U_1 V_{\theta 1} - U_2 V_{\theta 2})$, and enthalpy power as $Enthalpy\ Power = \dot{m}Cp(T_{01} - T_{02})$. For adiabatic flow these should agree; differences indicate either cooling effects or numerical precision issues.

EXAMPLES

EEE – 2 Stage HPT

The NASA Energy Efficient Engine (EEE) program was a joint effort between NASA and major engine manufacturers (General Electric and Pratt & Whitney) during the late 1970s and 1980s to develop advanced propulsion technologies for fuel-efficient commercial aircraft. The GE EEE high-pressure turbine (HPT) employs a two-stage configuration with moderate aerodynamic loading, selected through detailed system studies as the optimal approach for achieving the program's efficiency goals. Full-scale warm air turbine testing with simulated cooling flows demonstrated a design point efficiency of 90.0% (thermodynamic definition). Rolls-Royce had their own EEE turbine; however, theirs was single-stage. This turbine was not included in our verification because geometry and data were difficult to obtain.

The Tutorial on GitHub demonstrates how to go from the IGES files to extracting the curves for each blade profile, splitting the profile into suction and pressure sides along with the hub and shroud curve - splitting the suction and pressure sides which is needed by ADS (Aerodynamic Solutions, Inc.) [16] Mesh generation tool WAND to produce leading and trailing edge wedge mesh. The tutorial also demonstrates how to set up the case for solution using ADS. Figure 8 Top shows the data from the IGES. The blades are positioned correctly in the passage however they are not fitted to the passage streamlines. Figure 8 Bottom shows the fitted geometry. Figure 9 contains a plot of all the blade profiles

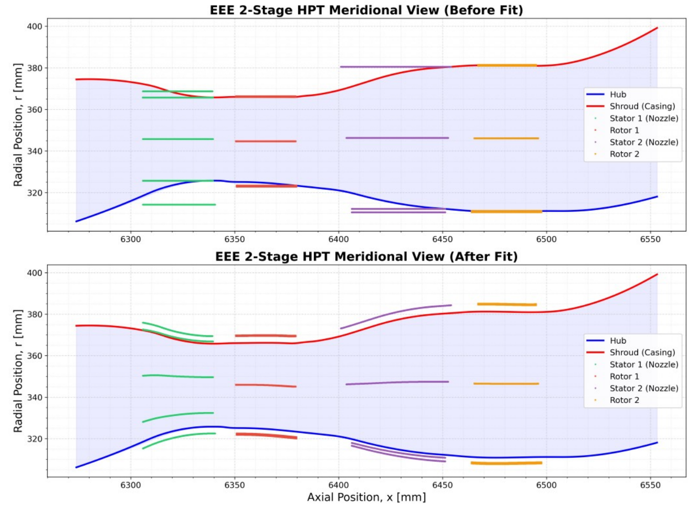

Figure 8. EEE-2 Stage HPT (GE). Top before fitting IGES profiles to passage streamlines; Bottom: After fitting IGES

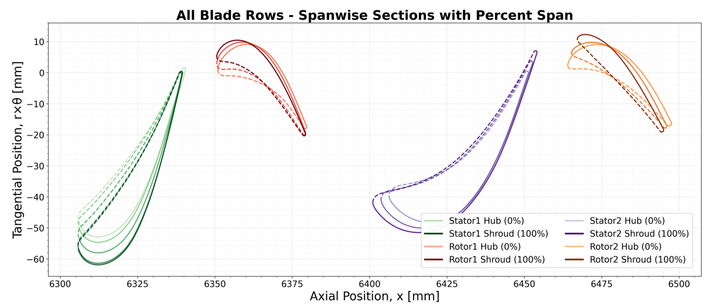

Figure 9. EEE All Blade Profiles from IGES data [17]

Table 1 contains the boundary conditions used to set up EEE simulation with ADS. These boundary conditions came from the data release from Claus et al. [17]. The mean Y+ for the simulation was below 1. Wilcox k-ω 98 turbulence model was used. The wall surfaces were treated as adiabatic. Steady state simulations were simulated for 30,000 iterations to ensure convergence. Figs. 10 and 11contain convergence plots in terms of mass flow and power, indicating the solution has converged. All simulation files are linked on the GitHub Tutorial.

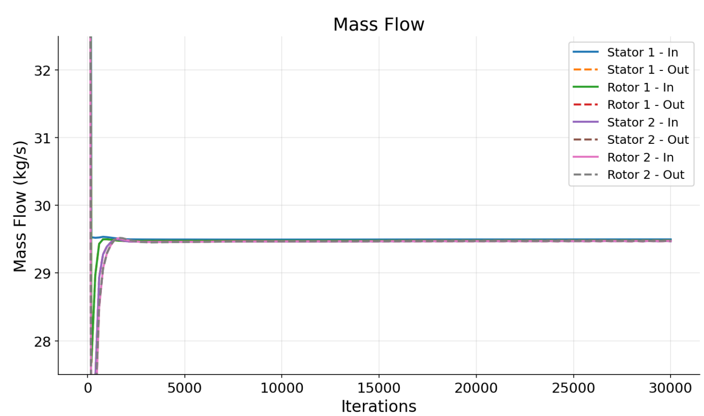

Figure 10. EEE MassFlow Convergence

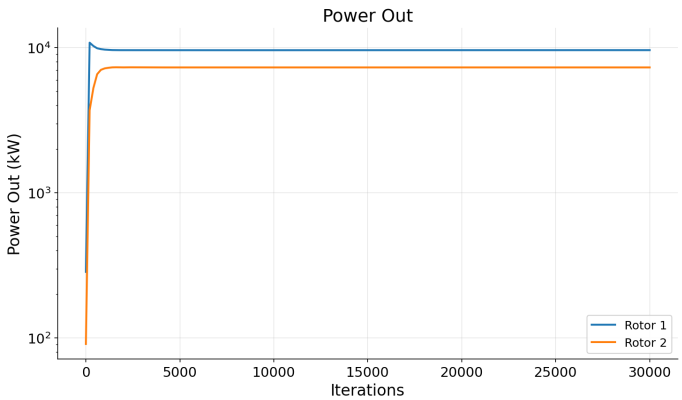

Figure 11. EEE ADS Simulation Power Convergence

Table 1. EEE Boundary Conditions

Boundary Conditions		
Inlet	P_0 [Bar]	12.57514
	T_0 [K]	1587
	M	0.2
Outlet	Ps [Bar]	2.303
Rotors	RPM	12400
Fluid	Air - variable Cp, γ	

EEE – CFD Simulation Results and Comparison

CFD was performed and evaluated prior to running Turbo Design. The results of the simulations are shown in Table 2. The equations to evaluate Efficiency, Stage Loading, Total Pressure loss are shown below as (Eqs 22-26). Turbo-Design currently does not implement accurate loss models so to compare; we calculate the Total Pressure Loss in CFD using (Eq 22) and apply it for each row in the setup.

$$P_{0,loss} = Y_p = \frac{P_{01R} - P_{02R}}{P_{01R} - P_2} \tag{22}$$

$$\eta = \frac{C_{p1}T_{01} - C_{p3}T_{03}}{C_{p1}T_{01} - C_{p3}T_{03,is}} \tag{23}$$

$$T_{03,is} = T_{01}\left(\frac{P_{03}}{P_{01}}\right)^{(\gamma-1)/\gamma} \tag{24}$$

$$\psi = \frac{H_{02} - H_{01}}{U^2} \tag{25}$$

Turbo-Design solves the radial equilibrium equations to balance mass flow between the rows by adjusting the static pressure between the rows while taking into account varying fluid properties. Figure 12 shows the convergence

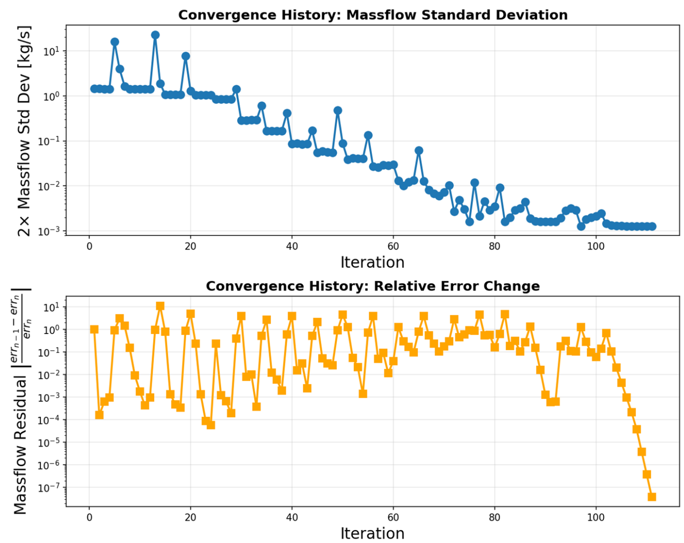

Figure 12. EEE: Turbo-Design Convergence History

Table 2. EEE: Simulation Results and Comparison with Turbo-Design

CFD Results	Stator 1	Rotor 1	Stator 2	Rotor 2
Efficiency		0.962		0.97
Stage Loading ψ		1.61		1.20
P_0 Loss (Yp)	0.057	0.088	0.069	0.014
Massflow [kg/s]	29.46			

The Table 3 is the results from Turbo-Design. The table columns are Stages 1 and 2 which consist of (Stator 1, Rotor 1) and (Stator 2, Rotor 2). The results of Table 3 are compared with Table 2 and the percent error is shown below. The comparison against CFD showed similar efficiency, loading, and mass flow. This demonstrates that the solver works for axial turbines provided that the loss modeling is accurate.

Table 3. EEE: Turbo-Design Results and % Error from CFD

Turbo-Design	Stage 1	Stage 2
Efficiency η	0.955	0.986
η % Error	0.7%	1.65%
Stage Loading ψ	1.56	1.26
ψ % Error	3.1%	5%
Massflow [kg/s]	29.4	
Massflow % Error	0.2%	

Radial Turbine CFD

The last term in the radial equilibrium equation ((Eq 15)) is often neglected in 1D analysis because it adds complexity. To test solver accuracy for radial machines where this term is significant, a radial turbine was developed internally at NASA using PyTurbo-Aero [18]. The geometry was exported to ADS for CFD simulation of a single rotor in the Relative Frame, then converted to the absolute frame using RPM. Pressure loss extracted from CFD was used in Turbo Design to compare predicted efficiency, Mach number, and mass flow.

Table 4 shows the boundary conditions used for the ADS simulation. Both CFD and Radial Equilibrium analyses assumed constant $\gamma = 1.35$. Absolute total pressure (P_0) and total temperature (T_0) from CFD were used as inputs to Turbo Design.

Table 4. Radial Turbine: CFD Boundary Conditions

Boundary Conditions		
Inlet	P_{0rel} [Bar]	50
	T_{0rel} [K]	1200
	M_{rel}	0.05
	β_2[Deg]	0
Outlet	Ps [Bar]	25
Rotors	RPM	-50000
γ (constant)	1.35	
Fluid	Air	

The CFD geometry included extended inlet and outlet regions to allow flow development, which are not present in Turbo Design. Additionally, Turbo Design requires a stator upstream of the rotor; a zero-loss stator was assumed with inlet flow angle $\alpha_1 = 0$. The stator exit angle α_2 was adjusted to match the rotor inlet flow angle from CFD. The rotor exit angle β_3 used in Turbo Design corresponds to the design blade angle used to construct the 3D geometry.

Radial Turbine CFD Results

ADS Simulation ran for 45000 iterations until the solution converged shown in Figure 13. Absolute flow conditions were converted from relative frame using PyTecplot and compared with the solution from Turbo-Design in Table 5. By applying the same total pressure loss coefficient (Yp = 0.136), Turbo Design matched mass flow within 2.2%, total-to-total efficiency within 0.8%, and total-to-static efficiency within 1.3%. Exit flow angles agreed within 1°, demonstrating that the radial equilibrium formulation correctly captures flow turning even when the meridional curvature term is significant. The 3.5% deviation in power is attributed to the compounding effect of small differences in mass flow and total temperature drop.

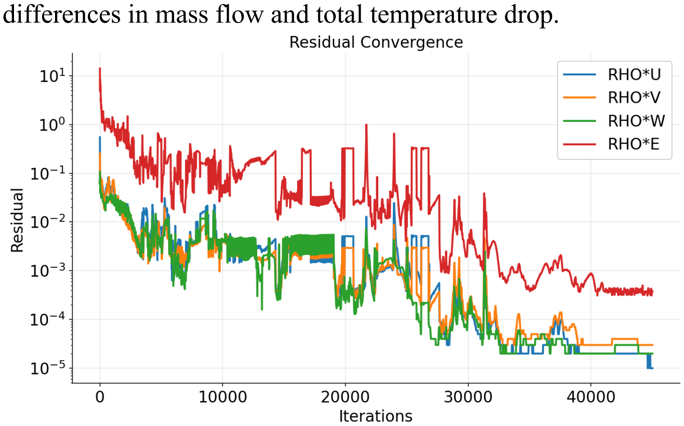

Figure 13. Radial Turbine ADS CFD Convergence

Table 5. Radial Turbine: Turbo-Design Comparison with CFD

	1D Model	CFD	% Difference
P0 Loss (Yp)	0.136	0.136	prescribed
Total-Total Efficiency	0.803	0.797	+0.8%
Total-Static Efficiency	0.338	0.334	+1.3%
Total Massflow (kg/s)	0.299	0.292	+2.2%
Enthalpy Power (kW)	23.23	22.45	+3.5%
Euler Power (kW)	23.23	22.46	+3.5%
T0 inlet (K)	1222.9	1222.9	0.0%
T0 exit mean (K)	1152.6	1174.1	-1.8%
P0 inlet (Pa)	536,657	536,657	0.0%
P0 exit mean (Pa)	427,458	428,240	-0.2%
beta1 avg (deg)	0.01	0.12	~0%
beta2 avg (deg)	51.2	51.7	-1.0%

CONCLUSION AND FUTURE WORK

This paper presented Turbo Design, an open-source Python-based radial equilibrium solver for turbomachinery analysis. The governing equations were derived from first principles, with particular attention to the meridional curvature terms required for radial machines. The solver was validated against CFD for the NASA EEE 2-stage HPT and an internally developed radial turbine, demonstrating agreement within 2% for mass flow and efficiency when supplied with accurate loss coefficients.

The team at NASA Glenn remains committed to advancing high-quality turbomachinery research and enabling the next generation of design tools. While the current Turbo Design tool offers a solid foundation, it lacks integrated loss models tailored to specific turbomachinery stages, and currently does not support compressor design — both axial and radial — which is a key area for future development.

One of the most impactful ways to drive improvement is through active collaboration with the broader research community. We invite universities, national laboratories, and industry partners to contribute optimization datasets, high-fidelity LES and DNS simulations, and experimental results. These shared resources will enable the development of advanced machine learning-based loss models that enhance Turbo Design's capabilities and accuracy. Join us in shaping the future of turbomachinery design through open collaboration and collective innovation.

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CODE INFORMATION

The code, tutorials, and CFD Simulation files are located at https://www.github.com/nasa/turbo-design

The authors acknowledge that the paper provides detailed derivations and descriptions. These can be better explained by exploring the documentation and tutorials in the repository. The repository will provide the latest description of the architecture and changes. If you find any issues or want any features, please submit a GitHub issue and your comments will be addressed.

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APPENDIX A
Entropy-Based Efficiency Derivation

The entropy-based efficiency (Eq 27b) shown below:

$$\eta_{TT} = \frac{w}{w + T_2 \Delta s} \quad (27b)$$

Is derived from the Gibbs equation for a steady-flow system (Eq 28a):

$$dh = Tds + vdP \quad (28a)$$

$$dh = Tds \ \ (at\ constant\ pressure) \quad (28b)$$

$$h_2 - h_{2s} \approx T_2 \Delta s \quad (28c)$$

When comparing two exit states at the same pressure P_2, the actual state (h_2, s_2) and the hypothetical isentropic state (h_{2s}, s_1) gives $dP = 0$ (Eq 28b). Integrating yields (Eq 28c), where T_2 approximates the temperature during dissipation. The quantity $(h_2 - h_{2s})$ represents the lost work. Since isentropic work is $w_{is} = w + (h_2 - h_{2s})$, efficiency becomes $\eta = \frac{w}{w+T_2\Delta s}$

The entropy generation Δs is computed from the relative frame analysis. For an ideal gas, the entropy differential is (Eq 29a):

$$ds = c_p \frac{dT}{T} - R\frac{dP}{P} \quad (29a)$$

$$h_{0R} = h + \frac{W^2}{2} = constant \quad (29b)$$

$$T_{0R,in} = T_{0R,out} \quad (29c)$$

$$\Delta s = Rln\left(\frac{P_{0R,in}}{P_{0R,out}}\right) \quad (29d)$$

In the rotating frame, the rotor appears stationary with no shaft work or heat transfer, so the relative total enthalpy h_{0R} is conserved (Eq 29b). For an ideal gas with constant c_p, this means the relative total temperature is also conserved (Eq 29c). Applying the integrated form of (Eq 29a) to relative total conditions, the temperature term vanishes since ln(1) = 0, yielding (Eq 29d). This is identical to (Eq 27c).

Solving the Radial Equilibrium Equation

(Eq 15) is the radial equilibrium equation – defined in the $\hat{\imath}_r$ frame of reference in its simplest form. Often *Vr* is assumed to be negligible for axial machines; however, when changes in hub and shroud alter mean radius, this term can be a source error if not considered.

$$\hat{\imath}_r : \frac{1}{\rho}\frac{dP}{dr} = \frac{V_T^2}{r} - \frac{V_M^2}{r_M}cos(\phi) - V_r\frac{dV_M}{dM} \quad (15)$$

Solving this (Eq 15) requires defining pressure in terms of meridional velocity. To accomplish this, pressure can be rewritten in terms of the ratio of total to static temperature ratio (Eqs 30-31).

$$T_0 = T + \frac{V^2}{2Cp} \qquad \frac{T}{T_0} = 1 - \frac{V^2}{2CpT_0} \quad (30)$$

$$\frac{P}{P_0} = \left(\frac{T}{T_0}\right)^{\frac{\gamma}{\gamma-1}} = \left(1 - \frac{V^2}{2CpT_0}\right)^{\frac{\gamma}{\gamma-1}} \quad (31)$$

(Eq 32) is the derivative of (Eq 31) with respect to the radius which shows how Static Pressure varies long the radius as the velocity changes.

$$\frac{dP}{dr} = \frac{d}{dr}\left[P_0\left(1 - \frac{V^2}{2CpT_0}\right)^{\frac{\gamma}{\gamma-1}}\right] \quad (32)$$

If we let $B = \left(1 - \frac{V^2}{2CpT_0}\right)^{\frac{\gamma}{\gamma-1}}$ and substituting V_M for V yields:

$$B(r) = \left(1 - \frac{V_M^2(1 + tan^2(\alpha))}{2CpT_0}\right)^{\frac{\gamma}{\gamma-1}} \quad (33)$$

To simplify the derivative, let C = $\frac{1+\tan^2(\alpha)}{2Cp}\frac{V_M^2}{T_0}$

The derivative of the right hand side of (Eq 32) can now be represented using variables *B*:

$$\frac{dP}{dr} = \frac{dP_0}{dr}B + P_0\frac{dB}{dr} \quad (34)$$

Where $\frac{dB}{dr}$ equals:

$$\frac{dB}{dr} = \frac{-\gamma}{\gamma - 1}(1 - C)^{\frac{1}{\gamma-1}}\frac{dC}{dr} \quad (35)$$

$$\frac{dC}{dr} = \frac{1 + tan^2(\alpha)}{2Cp}\left(\frac{2V_M}{T_0}\frac{dV_M}{dr} - \frac{V_M^2}{T_0^2}\frac{dT_0}{dr}\right) \quad (36)$$

All these substitutions can be used to rewrite the radial equilibrium equation in terms of Total Pressure (P_0), Meridional Velocity (V_M) and Total Temperature (T_0) as a function of radius. Substituting the derivative of B and C into (Eq 34) results in (Eq 37).

$$\begin{aligned}\frac{dP_0}{dr}B - \frac{P_0\gamma}{\gamma-1}(1-C)^{\frac{1}{\gamma-1}}\frac{(1+tan^2(\alpha))}{2Cp}\bigg(\frac{2V_M}{T_0}\frac{dV_M}{dr} \\ -\frac{V_M^2}{T_0^2}\frac{dT_0}{dr}\bigg) \\ = \rho\left(\frac{V_T^2}{r} - \frac{V_M^2}{r_M}\cos(\phi) - V_r\frac{dV_M}{dM}\right)\end{aligned} \quad (37)$$

A cleaner form can be obtained by setting:
$A = -\frac{P_0\gamma}{\gamma-1}(1-C)^{\frac{1}{\gamma-1}}\frac{(1+tan^2(\alpha))}{2Cp}$ which simplifies (Eq 37) to:

$$\frac{dV_M}{dr} = \frac{T_0}{2V_MA}\left(\rho\left(\frac{V_T^2}{r} - \frac{V_M^2}{r_M}cos(\phi) - V_r\frac{dV_M}{dM}\right) - B\frac{dP_0}{dr}\right) + \frac{V_M}{2T_0}\frac{dT_0}{dr} \quad (38)$$

V_r and V_T can both be expressed in terms of V_M allowing for V_M, T_0, and P_0 to be balanced radially.